Where K_{rot} represents the rotational kinetic energy expressed in terms of the body-fixed angular momentum vectors, K_{trans} is the translational kinetic energy, U is the potential energy which is a function of all the positions and dipole orientations. The vectors $\boldsymbol{\mu}_k$ can be taken to be unit vectors reflecting the dipole orientation, so they are subject to the constraint $\|\boldsymbol{\mu}_k\| \equiv 1$. The total dipole-dipole interaction potential then takes the form

$$\varphi_{ij}^{\text{dip}} = \varphi_{\text{LJ}}(r_{ij}) + \boldsymbol{\mu}_i^T \boldsymbol{L}(\boldsymbol{q}_{ij})\boldsymbol{\mu}_j,$$

with U given as the sum of these over pairs. The 3×3 matrix $\boldsymbol{L}$ is given by

$$\boldsymbol{L}(\boldsymbol{q}_{ij}) = \frac{1}{r_{ij}^3} - \frac{3}{r_{ij}^5}\boldsymbol{q}_{ij}\boldsymbol{q}_{ij}^T, \qquad \boldsymbol{q}_{ij} = \boldsymbol{q}_i - \boldsymbol{q}_j.$$

The rotational kinetic energy may be written

$$K_{\text{rot}}(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \ldots, \boldsymbol{\pi}_{\text{N}}) = \frac{1}{2}\sum_{i=1}^{\text{N}} \frac{\boldsymbol{\pi}_{i,1}^2}{I_1} + \frac{\boldsymbol{\pi}_{i,2}^2}{I_2} + \frac{\boldsymbol{\pi}_{i,3}^2}{I_3},$$

with I_1, I_2, I_3 the moments of inertia, taken to be diagonal with respect to a suitable coordinate system. Introducing a rotation matrix $\boldsymbol{\Theta}_k$ for the kth dipole, we have $\boldsymbol{\mu}_k(t) = \boldsymbol{\Theta}_k\boldsymbol{\mu}_k(0)$, where $\boldsymbol{\mu}_k(0)$ is a reference configuration of the dipole (its initial state). Then

$$K_{\text{rot}} = \sum_{i=1}^{\text{N}} \text{Tr}\left(\dot{\boldsymbol{\Theta}}_i \boldsymbol{R} \dot{\boldsymbol{\Theta}}_i^T\right),$$

with $\boldsymbol{R}$ defined in terms of $\boldsymbol{T} = \text{diag}(I_1, I_2, I_3)$ by (4.40).

We now have two alternatives for representing the rotational motion. In the first, the rotation matrix describing the rigid body's orientation is the dynamical quantity (and we need to enforce the orthogonality of this matrix as a constraint). This leads to a constrained integration method, which makes use of SHAKE (in a somewhat different way from the method [84] which constrains the relative positions of particles). In the second formulation, the orthogonality is used to perform a reduction to an angular velocity (or momentum) representation with respect to the body frame of reference.

A Hamiltonian description of the motion may be formulated in either system of notation. The Hamiltonian formulation with rotation matrices involves variables $\boldsymbol{q}_{\text{cm}}, \boldsymbol{p}_{\text{cm}}$ (phase space variables describing the center of mass motion) and $\boldsymbol{\Theta}, \boldsymbol{\Pi}$, which are 3×3 matrices. The Hamiltonian is (for a single rigid body):

$$H_{\text{RB}} = \frac{\|\boldsymbol{p}_{\text{cm}}\|^2}{2M} + \frac{\text{Tr}\left(\boldsymbol{\Pi}\boldsymbol{R}^{-1}\boldsymbol{\Pi}^T\right)}{2} + U(\boldsymbol{q}_{\text{cm}}, \boldsymbol{\Theta}),$$